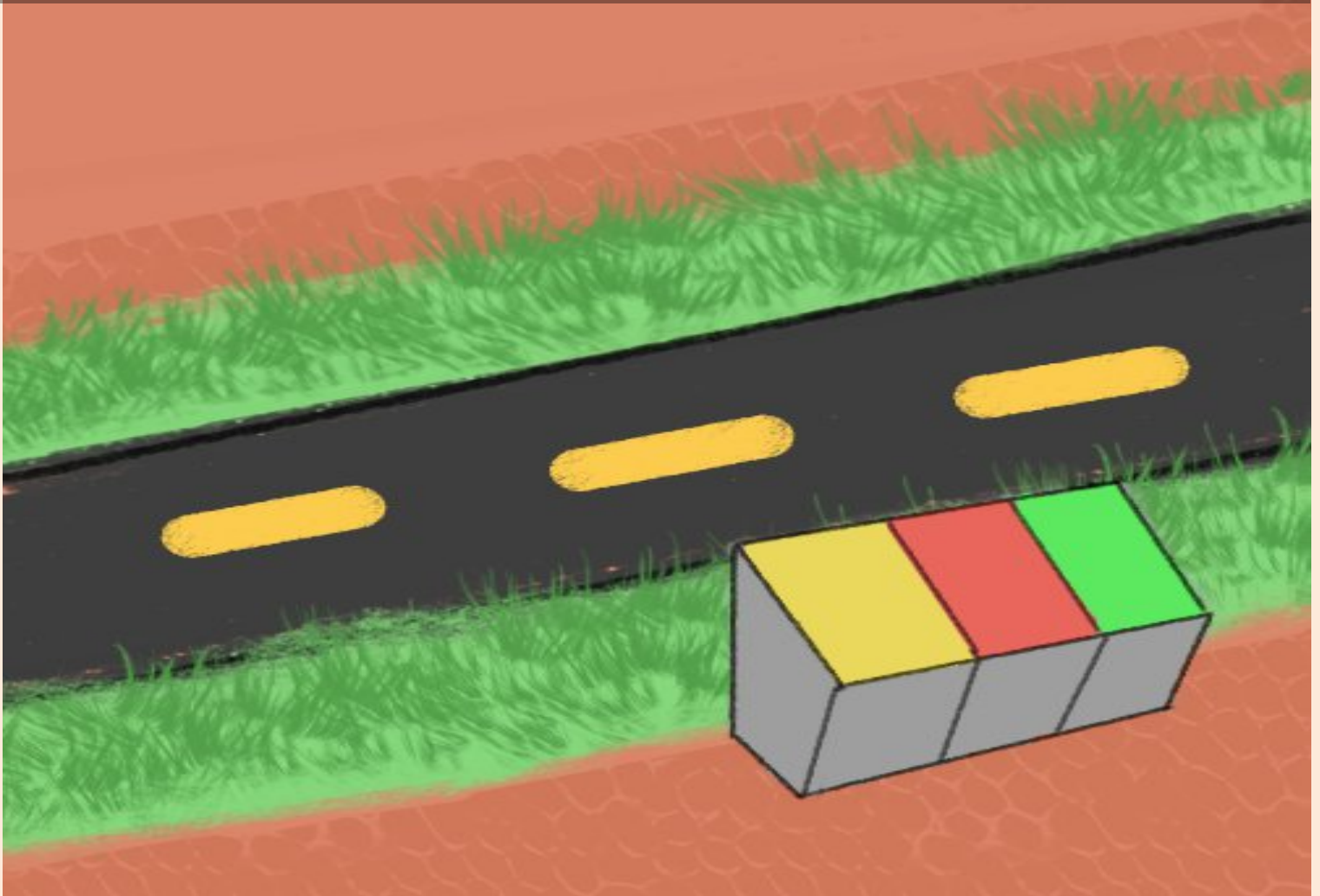


Project Proposal



Smart Street Section Simulation

A smart drainage and waste water management system for cities.

Team Members

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1. Problem Statement

Rapidly expanding urban centers in developing nations, particularly across Sub-Saharan Africa, face compounding civil engineering challenges. Inadequate waste management infrastructure leads to heavy street littering; uncollected solid waste then clogs open drainage channels during severe rainy seasons, causing severe localized flooding, public health hazards, and rapid degradation of urban roads.¹

A study of flood risk drivers in the Greater Kumasi Metropolitan Area, Ghana, found that poor waste disposal into drainage channels is one of the principal causes of urban flooding², and city authorities in Freetown, Sierra Leone draw direct links between poor waste management, clogging of drains, and subsequent flooding.³

Traditional open concrete drainage channels invite illegal dumping and lack both the capacity and the automated controls needed to manage peak storm runoff combined with heavy debris loads. The result is a recurring cycle: waste clogs drains, drains overflow, floods damage roads and property, and the cost of recovery diverts resources that could have maintained the waste and drainage systems in the first place.

2. Proposed Solution

In direct support of United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities), and specifically its targets on reducing deaths and economic losses from water-related disasters⁴, we propose developing a **smart urban street section simulation**: a software model of a hybrid green-gray (nature-integrated plus engineered) street drainage and waste system designed for high-density tropical urban environments. Green-gray combinations of permeable surfaces, bio-retention cells, and conventional drainage are an actively researched approach to sustainable urban storm water management⁵, and full-scale streets combining permeable pavement with bio-retention facilities have demonstrated measurable peak-flow reductions in practice.

The simulation integrates four automated sub-systems:

1. **Permeable Bio-Retention Surface:** models soil moisture saturation to control automated actuator gates that divert excess storm runoff into sub-surface channels during heavy rainfall.
2. **Automated Debris Filtration & Flushing:** simulates smart gutter filters within the channel that detect debris blockages and trigger hydraulic flushing.

¹D. Arinabo, "Understanding the Evolving Nature of Urban Flood Risks in Sub-Saharan Africa: The Case of Kampala City, Uganda," IntechOpen, 2024, <https://www.intechopen.com/chapters/1179231>.

²K. B. Enu, A. Zingraff-Hamed, D. O. Appiah, and S. Pauleit, "Narrowing down the drivers of flood risk in medium-sized sub-Saharan African cities: insights from the Greater Kumasi Metropolitan Area, Ghana," Hydrological Sciences Journal, vol. 69, no. 14, pp. 1897–1912, 2024, <https://doi.org/10.1080/02626667.2024.2401605>.

³CDP, "Water, Waste and Adaptation," 2020, <https://www.cdp.net/en/insights/water-waste-and-adaptation>.

⁴United Nations, "Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable," UN Sustainable Development Goals, <https://sdgs.un.org/goals/goal11>.

⁵H. Tansar, F. Li, F. Zheng, and H.-F. Duan, "A critical review on optimization and implementation of green-grey infrastructures for sustainable urban stormwater management," AQUA — Water Infrastructure, Ecosystems and Society, vol. 73, no. 6, 2024, <https://iwaponline.com/aqua/article/73/6/1135/102593/>.

3. **Underground Smart Waste Reservoirs:** models subterranean high-capacity waste bins (inspired by deep-collection systems such as Moloch's [8] semi-underground containers⁶) receiving diverted debris and general pedestrian waste.
4. **IoT Telemetry & Municipal Alert Network:** processes simulated sensor inputs across the street section to generate real-time municipal alerts for bin collection and mechanical maintenance before flooding occurs.

Justification for a Computer Program being the best solution to this problem

Testing complex, multi-actuator civil infrastructure featuring bio-retention soils and subterranean waste storage is cost-prohibitive to build physically without prior validation. A software simulation provides a dynamic computational test bed to model system behavior under various rainfall intensities and waste accumulation rates, demonstrating automated flood prevention and waste routing in a controlled, risk-free environment.

3. Expected Outcomes

Upon completion, the project will deliver the following concrete outcomes:

- A working, interactive 3D simulation of a representative urban street section, demonstrating automated flood prevention under design storms ranging from light drizzle to tropical downpour.
- Demonstrable automation logic: actuator gates opening at soil saturation above 80%, filter flushing at flow reduction above 35%, pickup requests at 85% bin fill, and overflow protection at 100%.
- A live municipal alert feed proving the telemetry and dispatch concept, with red/yellow/green indicators per street section.
- Quantified performance metrics (e.g., peak flow reduction versus a conventional concrete channel baseline) benchmarked against MATLAB reference calculations.
- A maintainable, documented code base on GitHub with weekly development logs, sprint milestones, and a final technical report.
- A predictive analytics module that estimates time-to-overflow and issues early flood warnings before channel capacity is breached. (This is a stretch goal, we will ask TAs for guidance)

4. System Design

Module 1: Bio-Retention Surface & Automated Actuator Control

- **Inputs:** simulated rainfall intensity (mm/hr), surface runoff rate (L/s), and soil moisture saturation percentage (0–100%, from simulated soil moisture sensors).

- **Processing:** the C++ backend evaluates saturation against preset criteria (saturation above 80% triggers actuation), including cooldown-timer logic so gates do not remain open after rain stops while soil dries slowly.
- **Outputs:** dynamic 3D state change in which channel actuator gates open and close based on telemetry, redirecting excess water into subterranean channels.

Module 2: Channel Hydraulics & Filtration Flushing Mechanism

- **Inputs:** debris accumulation volume (kg or particle count) and the differential water flow rate before and after the filter mesh ($\Delta Q = Q_{in} - Q_{out}$).
- **Processing:** computes flow-restriction ratios. When ΔQ exceeds a critical threshold (greater than 35% flow reduction), a hydraulic flush event is triggered. If ΔQ remains high post-flush, the system enters a “Filter Clogged / Maintenance Required” fault state.
- **Outputs:** 3D animation of debris diversion into the subterranean repository and updated fluid flow rates.

Module 3: Subterranean Smart Bin Monitoring & Alert Dispatch

- **Inputs:** real-time mass load (kg) and ultrasonic fill-level height (cm) inside the subterranean reservoir.
- **Processing:** calculates volume capacity. At 85% fill, the software triggers an automated municipal pickup request; at 100% it flags an overflow warning and halts further debris routing to that bin.
- **Outputs:** visual fill-level gauge on the Godot HUD and simulated JSON/log-based municipal dispatch alerts.

Module 4: System Maintenance State & Emergency Override

- **Inputs:** manual admin toggles or automated fault flags (e.g., persistent filter blockage, actuator mechanical failure).
- **Processing:** overrides automated sensor loops, locking actuator gates in a safe failure state (fully open, to prevent flooding) and flagging maintenance personnel on the dashboard.
- **Outputs:** visual status indicators (red/yellow/green health states per street section) and event log outputs.

Stretch Goal: Predictive Flood Risk Analytics

- **Inputs:** historical rainfall trends combined with live sensor rate-of-change data (d/dt of soil moisture and channel head levels).
- **Processing:** time-series extrapolation to estimate time-to-overflow.
- **Outputs:** early-warning alert predicting flash-flood risk N minutes before channel capacity is breached.

5. Planned Use / Test Cases

The simulation will be validated against the following test cases.

ID	Scenario / Input	Expected Behavior	Module
TC-01	Light rain (2 mm/hr); soil moisture 40%	Actuator gates remain closed; runoff infiltrates through the bio-retention surface; no alerts raised.	M1
TC-02	Heavy rain (60 mm/hr) sustained until soil moisture exceeds 80%	Actuator gates open and excess runoff is diverted to the subterranean channel; gates close only after moisture falls below threshold and the cooldown timer expires.	M1
TC-03	Debris load causing < 35% flow reduction across the filter mesh	No flush triggered; filter operates normally; flow rates update on the HUD.	M2
TC-04	Debris load causing > 35% flow reduction ($\Delta Q/Q_{in} > 0.35$)	Hydraulic flush event is triggered; flow rates recover; diverted debris is routed to the underground reservoir.	M2
TC-05	ΔQ remains > 35% after a completed flush	System enters the "Filter Clogged / Maintenance Required" fault state; maintenance flag raised on the dashboard.	M2, M4
TC-06	Bin mass load or ultrasonic fill level reaches 85% capacity	Automated municipal pickup request is dispatched (JSON/log alert); fill-level gauge updates on the HUD.	M3
TC-07	Bin fill level reaches 100% capacity	Overflow warning issued; further debris routing to that bin is halted.	M3
TC-08	Manual admin toggle or simulated actuator mechanical failure	Emergency override engages: gates lock fully open (fail-safe against flooding); red status indicator shown; event logged.	M4
TC-09	Stretch: sustained rise in soil-moisture rate-of-change (d/dt) under continued rainfall	Predictive module estimates time-to-overflow (T_{overflow}) and issues an early-warning alert before channel capacity is breached.	Stretch

6. Implementation Plan

The proposed application employs a decoupled, multi-tier software architecture designed to balance computational rigor with real-time visual demonstration:

- **C++ Core** : the primary computational backend, executing the physical simulation models (rational-method runoff calculations, soil saturation logic, and debris-driven flow restrictions) and managing the IoT sensor state machine.
- **Godot 4 Engine & GDScript**: the frontend visualization layer. GDScript orchestrates UI dashboards, camera controls, and visual signal bindings, receiving processed state data from the C++ backend via GDExtension.
- **MATLAB**: used offline to benchmark the hydraulic differential equations and calibrate parameter constants prior to C++ implementation.
- **Blender & 3D Asset Pipeline**: custom structural components (subterranean drainage channels, filtration meshes, smart bin reservoirs) are modeled in Blender, while standard urban environment props are integrated from open-access 3D repositories to prioritize core software development.

7. Team Responsibilities and Timelines

Project Management, Collaboration, and Documentation

To ensure accountability, equal contribution, and a smooth team workflow:

- **Version control**: GitHub will store all codebase iterations, manage feature branches and track individual code contributions.
- **Automated time tracking**: team members will use WakaTime integrated into their code editors to record hours spent programming on the project.
- **Development logs & agile milestones**: tasks will be assigned on a weekly sprint cycle. After each work session, members submit a brief “Devlog” entry alongside their GitHub commits.
- **Weekly progress reports**: weekly logs, automated time logs, and completed tasks will be compiled into a continuous progress report, directly supporting milestone submissions and the final technical write-up.

Team Members

- **Benie Gouli (Visualization & Frontend)**: Godot 4 frontend (HUD dashboards, camera controls, 3D state visualization), Module 3–4 status and alert UI, and the Blender asset pipeline.
- **Ian Mwenda (Hydraulics & Analytics)**: Module 2 (channel hydraulics and filtration/flushing logic), MATLAB benchmarking and parameter calibration, and the predictive flood-risk stretch goal.
- **Chelsea Chepkoech (C++ Core Engine)**: (soil saturation model and actuator state machine), rational-method runoff calculations, and GDExtension bindings between the C++ backend and Godot.

References

The footnotes throughout this document correspond to the following full references:

- [1] D. Arinabo, “Understanding the Evolving Nature of Urban Flood Risks in Sub-Saharan Africa: The Case of Kampala City, Uganda,” IntechOpen, 2024. <https://www.intechopen.com/chapters/1179231>
- [2] K. B. Enu, A. Zingraff-Hamed, D. O. Appiah, and S. Pauleit, “Narrowing down the drivers of flood risk in medium-sized sub-Saharan African cities: insights from the Greater Kumasi Metropolitan Area, Ghana,” Hydrological Sciences Journal, vol. 69, no. 14, pp. 1897–1912, 2024. <https://doi.org/10.1080/02626667.2024.2401605>
- [3] CDP, “Water, Waste and Adaptation,” 2020. <https://www.cdp.net/en/insights/water-waste-and-adaptation>
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- [10] M. Siddique, T. Ahmed, and M. S. Husain, “Flood Monitoring and Early Warning Systems – An IoT Based Perspective,” EAI Endorsed Transactions on Internet of Things, vol. 9, no. 2, e4, 2023. <https://doi.org/10.4108/eetiot.v9i2.2968>

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- Research (the sources were cross-checked)
- Test-cases polishing
- Grammatical correction

This project is our own original work and AI was not used in the brainstorming of the idea.